Muhammad Hassan Haider

Full Stack AI/LLM & Backend Developer

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Summary — Full Stack AI/LLM and Backend Developer with 5+ years of experience shipping end-to-end products: scalable Django/FastAPI backends, React/Next.js + TypeScript frontends, and production-grade RAG, agentic AI, and LLM fine-tuning (LoRA/DoRA) systems built with LangChain, LlamaIndex, ChromaDB, and Ollama/vLLM. Uses Claude Code and AI-assisted workflows to move quickly from API design to polished UI and Dockerized deployment. Delivered systems cutting manual effort by 70–90% across ETL automation, payments, and document intelligence. Seeking Full Stack AI/LLM Engineer, Generative AI, or Backend+AI roles.

Skills

- **Languages**
- Python, SQL
- TypeScript, JavaScript
- **Backend**
- Django & DRF
- FastAPI, Flask
- WebSockets, REST APIs
- Celery & Redis
- **AI / LLM**
- RAG & Agentic AI
- LangChain, LlamaIndex
- Fine-Tuning (LoRA/DoRA)
- PEFT / Transformers
- Prompt Engineering
- ChromaDB, FAISS
- vLLM / Ollama Inference
- **Frontend**
- React, Next.js
- Tailwind CSS
- Streamlit
- **ML / Data**
- TensorFlow / Keras
- spaCy, Scikit-Learn
- NumPy & Pandas
- **Cloud / DevOps**
- AWS (S3, Lambda, Glue)
- Docker & Compose
- PostgreSQL, Snowflake
- **AI Dev Tools**
- Claude Code
- MCP Servers
- Git & GitHub

Experience

Obsidian Realm Technologies (Guangzhou, China) **March 2026 – July 2026**
AI Engineer

- Designed and deployed LLM-based RAG and agentic AI solutions using Python, FastAPI, LangChain, and LlamaIndex
- Built document processing, chunking, embedding, and retrieval pipelines for knowledge-based AI applications
- Deployed local/open-source models via Ollama and vLLM, optimizing GPU utilization and inference performance
- Developed backend APIs integrating AI models with existing applications, databases, and third-party services

QubitDance (Chengdu, China) **June 2025 – January 2026**
Backend Developer

- Developed microservices architecture with Django and DRF for scalable backend solutions
- Designed RESTful APIs and containerized services with Docker for portable environments
- Implemented JWT authentication, role-based access control, and query optimization to reduce latency

Techmate Technologies **October 2022 – June 2025**
Python Developer

- Automated ETL pipelines using Python/Pandas, integrated with AWS and Snowflake
- Implemented Stripe/PayPal payment gateways for e-commerce solutions
- Built Streamlit dashboards with real-time data sync and geospatial validation via HereAPI

Code Embeded **November 2021 – October 2022**
Python Django Developer

- Developed scalable Django web applications with RESTful APIs using DRF
- Implemented JWT authentication and role-based access control systems

Education

University of Gujrat (Hafiz Hayat Campus) **September 2016 – October 2020**
BS in Software Engineering

Languages

- **Urdu** (Native) — **Punjabi** (Native) — **English** (Professional) — **Chinese** (HSK Level 1)

Projects

VoiceForge – Local Multi-User AI Voice and Video Studio
Technologies: python, fastapi, postgresql, websockets, react, typescript, tailwind, ollama, onnx runtime, ffmpeg

- Built a fully offline, multi-user AI studio for streaming voice chat, voice cloning and AI video generation, running entirely on local models (Ollama, Whisper, Pocket TTS) on CPU
- Streamed LLM replies over WebSockets with sentence-level TTS, so audio starts before the full reply is written, and added a Postgres job queue (SKIP LOCKED + LISTEN/NOTIFY) for heavy video jobs without needing Redis
- Built AI video pipelines for narrated explainer videos with Whisper-aligned captions and Wav2Lip lip-synced avatars, with JWT auth, a React + TypeScript frontend and Docker Compose deployment

PlotForge – Scientific Chart Generation Platform (Full Stack)
Technologies: python, django, django rest framework, matplotlib, plotly, pandas, scipy, stripe, docker, next.js, react, typescript, tailwind, tanstack query, zustand

- Built a full-stack SaaS platform where researchers paste or upload data and get publication-quality charts from 390+ chart types (volcano plots, Kaplan-Meier curves, heatmaps, phylogenetic trees), with journal color sets and figure sizes in inches
- Designed a plugin-based chart registry in Django where each chart is one class, so one API endpoint validates, charges credits, renders and auto-refunds on failure, backed by an idempotent credit wallet, Stripe/PayPal checkout and an admin area
- Built the Next.js + TypeScript frontend with a searchable chart gallery, a chart editor with demo data and live settings, JWT auth with Google sign-in and token refresh, light/dark themes and a public landing page

GuardPipe – Open-Source AI Guardrails Middleware for LLMs **PyPI | Live Demo**
Technologies: python, pydantic, fastapi, langchain, llamaindex, openai/anthropic sdk, gradio

- Published `guardpipe` on PyPI – middleware detecting prompt injection, jailbreaks, PII, and toxic content in LLM I/O with one-line integration for OpenAI, LangChain, LlamaIndex, FastAPI, and Django
- Added RBAC, rate limiting, and audit logging, plus a live Hugging Face Space demo for real-time testing

AgentForge – Multi-Tenant Agentic RAG Platform
Technologies: python, django, drf, llamaindex, ollama, chromadb, celery, redis, docker

- Built a multi-tenant platform where users create AI agents backed by private PDF knowledge bases, with tenant isolation and a JWT-secured REST API
- Implemented a LlamaIndex ReActAgent with autonomous retrieval decisions and Celery/Redis-based incremental background ingestion, fully containerized with a local LLM stack (zero cloud dependency)

LLM Fine-Tuning Pipeline (LoRA / DoRA)
Technologies: python, peft, transformers, qwen3, gguf, ollama, pytorch

- Built an end-to-end fine-tuning pipeline for Qwen3-4B-Instruct on custom PDFs, covering data prep, LoRA/DoRA training, model merging, and GGUF export
- Engineered a PDF-to-training-data conversion script and validated inference quality locally via Ollama, completing a full fine-tune-to-deployment loop

Real-Time Voice AI System (ASR → LLM → TTS)
Technologies: python, sensevoice, ollama, vllm, indextts, websockets, fastapi, docker

- Built a production voice AI pipeline on a Tesla V100S server integrating SenseVoice ASR, Ollama/Qwen3 LLM inference, and IndexTTS via vLLM for speech synthesis
- Built a FastAPI orchestration layer with SQLite-backed memory and sentence-level chunked streaming for low-latency real-time conversations

PixelRAG Hybrid – Visual Retrieval-Augmented Generation System
Technologies: python, llamaindex, ollama, llava, chromadb, faiss, sqlite

- Built a Visual RAG system rendering web pages as screenshots (via CDP Headless Chrome) instead of HTML parsing, eliminating 40%+ content loss from traditional scrapers
- Used LLaVA:7B for screenshot transcription and Sentence Window chunking with dual ChromaDB/FAISS support, fully local with zero cloud API calls

RAGForge – Multi-Agent Local RAG Framework
Technologies: python, llamaindex, ollama, chromadb, faiss, bm25, ragas

- Architected a fully offline multi-agent RAG framework with LlamaIndex Workflows, implementing 12 chunking and 12 retrieval strategies (Hybrid, RRF, Auto-Merging, Multi-Query)
- Built CLI-configurable pipelines with automatic index caching and integrated RAGAS evaluation for offline benchmarking

AsifMolBio Lead Platform **November 2024 – Present**
Technologies: python, django, drf, postgresql, paypal, stripe, aws s3

- Developed a Django/DRF backend with JWT auth, dual PayPal/Stripe payment integration, and AWS S3 media storage
- Optimized PostgreSQL queries via Django ORM (40% faster API response) and built an admin dashboard for full CRUD content control

ETC Dashboard **December 2024 – June 2025**
Technologies: python, streamlit, snowflake, aws s3, aggrid, jwt

- Built a secure Streamlit dashboard with JWT auth and RBAC as the frontend for real-time ETL-processed data visualization
- Designed Snowflake-integrated ETL pipelines and interactive AgGrid tables, reducing manual data handling by 90%

X-NFT Microservice Platform
Technologies: python, django, drf, postgresql, pydantic

- Built a Django/DRF microservice for NFT and Collection management, integrated with a Solidity blockchain service for mint, buy, transfer, and burn operations
- Implemented multi-tenant architecture with RBAC and advanced search/filter/sort for NFTs and Collections, similar to OpenSea

PayUni – E-commerce Payment Microservice
Technologies: python, django, drf, postgresql, jwt

- Developed a multi-tenant e-commerce microservice with dynamic role management, enabling merchants to list and manage products with custom user roles
- Integrated crypto and fiat payment support via a separate payment microservice, with full exception handling for payment workflows

APIFY – API Marketplace Platform
Technologies: python, django, drf, alipay, wechat pay, postgresql, jwt

- Built an API marketplace platform (similar to RapidAPI) with API/version/endpoint/pricing management and demand-and-offer purchasing
- Implemented group purchase for shared API subscriptions and integrated Alipay/WeChat Pay through an existing payment microservice

M-Bizz (AI Marketing Bots) **July 2024 – November 2024**
Technologies: python, django, langchain, openai, rag, pandas, firebase, postgresql

- Built conversational AI chatbots using LangChain and OpenAI GPT models on a scalable FastAPI backend for business workflow integration
- Implemented NLP preprocessing pipelines (Pandas) and Retrieval-Augmented Generation (RAG) for domain-specific knowledge

TaskFlowAI: Task Management Platform **June 2025 – Present**
Technologies: python, django, postgresql, jwt, langchain, RAG

- Developed a Django backend for a task management system with modular apps (accounts, project, task) and a custom user model with role-based access
- Optimized PostgreSQL performance with indexes and `select_related` queries, with role-based permissions restricting project/task operations

MD-EXAM Pharmacy System **April 2023 – July 2023**
Technologies: python, django, mongodb, djongo, shipstation api, pymongo

- Developed a Django pharmacy management system using Djongo/PyMongo for MongoDB integration, with role-based access (Super Admin, Staff)
- Integrated the ShipStation API for automated label generation and built a daily sync command to auto-update order statuses

Certifications

- Python Complete Course (Udemy) — Python & Django Framework (Udemy) — SQL & MySQL (Udemy) — Web Scraping with BeautifulSoup (Udemy)